

New chat



L6_S1_A.pdf
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Task: Generate a lecture scribe.

Purpose: A lecture scribe is a faithful, exam-ready reconstruction of what was taught in class. The scribe should function as reference material suitable for preparing for a reading-based / closed-notes exam.

The context is fixed and authoritative.

Use ONLY the provided lecture slides. The lecture scribe guidelines govern how the scribe must be written.

Everything included must be traceable to the provided context.

Do not use external knowledge or introduce material not present in the context.

Prompting Style: PROMPTING STYLE 3: CHAIN-OF-THOUGHT (CoT) PROMPTING.

Chain-of-Thought Requirements:

While writing the lecture scribe, reason step by step.

Reasoning may appear in the final scribe to improve clarity for exam preparation.

All reasoning must come only from the provided context.

Content Requirements:

The lecture scribe should allow a student to:

- Recall definitions and notation
- Reconstruct proofs and derivations
- Follow worked examples step by step
- Understand logical dependencies between concepts

Accordingly, include:

- Definitions and notation
- Assumptions and conditions

- Statements of theorems or results
- Proofs or proof sketches, step by step, exactly as presented
- Worked examples with intermediate steps, exactly as shown in the lecture

Do NOT include:

- Creative explanations
- Simplified tutorials
- New intuition
- New examples
- Material not discussed in the lecture or shown in the slides

Explanation Constraints are - Explain definitions and assumptions explicitly. Require step-by-step proofs, derivations, and example solutions when they appear in the lecture. Do not provide alternative reasoning paths or additional explanations.

Structural Requirements: Follow the structure and flow of the lecture as presented in the slides. Organize the scribe clearly with appropriate sections and subsections so that it can be used reliably for revision.

Tone: Maintain a formal, academic tone suitable for exam revision. Prioritize correctness and faithfulness to the lecture. Do not invent content.

Finally:

Write the lecture scribe such that a student studying only this document can reliably revise exactly what was taught in the lecture.

Below is a **lecture scribe** reconstructed **strictly and only** from the provided lecture slides for **CSE400 – Lecture 6: Discrete Random Variables, Expectation and Problem Solving**.

The structure, notation, examples, and derivations follow the **exact flow of the**

lecture slides, and all content is traceable to the slides provided.

This document is written to be **exam-ready** for a **closed-notes / reading-based exam**.

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation and Problem Solving

Instructor: Dhaval Patel, PhD

Date: January 22, 2025

1. Random Variables (RVs)

1.1 Motivation and Concept

A **random variable (RV)** is a function that maps outcomes of a random experiment (sample space) to numerical values.

- The **distribution** of a random variable can be visualized using a **bar diagram**.
- The **x-axis** represents the values that the random variable can take.
- The **height of the bar** at value a_i represents:

$$\Pr[X = a_i]$$

- Each probability is computed by finding the probability of the **corresponding event** in the sample space.
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2. Types of Random Variables

2.1 Discrete Random Variables

A **discrete random variable** has the following properties:

- **Countable support**
- Defined using a **probability mass function (PMF)**
- Probabilities are assigned to **single values**
- Each possible value has **strictly positive probability**

2.2 Continuous Random Variables

A **continuous random variable** has the following properties:

- **Uncountable support**
 - Defined using a **probability density function (PDF)**
 - Probabilities are assigned to **intervals of values**
 - Each specific value has **zero probability**
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3. Example: Random Variable from Coin Tossing

Example 1: Tossing 3 Fair Coins

Experiment: Toss 3 fair coins.

Let Y denote the **number of heads** observed.

Possible Values of Y :

$$Y \in \{0, 1, 2, 3\}$$

Probability Computation:

- $P(Y = 0) = P(t, t, t) = \frac{1}{8}$
- $P(Y = 1) = P(t, t, h), (t, h, t), (h, t, t) = \frac{3}{8}$
- $P(Y = 2) = P(h, h, t), (h, t, h), (t, h, h) = \frac{3}{8}$
- $P(Y = 3) = P(h, h, h) = \frac{1}{8}$

Verification of Total Probability:

Since Y must take one of the values 0 through 3,

$$1 = P\left(\bigcup_{i=0}^3 \{Y = i\}\right) = \sum_{i=0}^3 P(Y = i)$$

4. Probability Mass Function (PMF)

4.1 Definition

A random variable that can take **at most a countable number of values** is called **discrete**.

Let X be a discrete random variable with range:

$$R_X = \{x_1, x_2, x_3, \dots\}$$

where the range is **finite or countably infinite**.

The function:

$$P_X(x_k) = P(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X .

4.2 PMF Property

Since X must take **one of the values** x_k , we must have:

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

5. PMF Example: Given Functional Form

Example

The PMF of a random variable X is given by:

$$p(i) = c \frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

where $\lambda > 0$.

Tasks:

1. Find $P(X = 0)$
 2. Find $P(X > 2)$
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Step 1: Determine Constant C

Since probabilities must sum to 1:

$$\sum_{i=0}^{\infty} p(i) = 1$$

Substitute:

$$c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using the known series:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

We get:

$$ce^{\lambda} = 1 \quad \Rightarrow \quad c = e^{-\lambda}$$

Step 2: Compute $P(X = 0)$

$$P(X = 0) = e^{-\lambda} \frac{\lambda^0}{0!} = e^{-\lambda}$$

Step 3: Compute $P(X > 2)$

$$\begin{aligned} P(X > 2) &= 1 - P(X \leq 2) \\ &= 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - (e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2} e^{-\lambda}) \end{aligned}$$

6. Bayes' Theorem (Recap)

6.1 Conditional Probability Identity

Using:

$$\Pr(A \mid B) = \Pr(B \mid A) \Pr(A)$$

We obtain:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i) \Pr(B_i)}{\sum_{j=1}^n \Pr(A \mid B_j) \Pr(B_j)}$$

This is known as **Bayes' Formula (Proposition 3.1)**.

6.2 Terminology

- $\Pr(B_i)$: A priori probability
(probability formed from self-evident or presupposed models)
 - $\Pr(B_i \mid A)$: Posterior probability
(probability computed after observing event A)
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7. Bayes' Theorem – Example 1: Auditorium with 30 Rows

Problem Setup

- An auditorium has **30 rows**
- Row 1 has **11 seats**
- Row 2 has **12 seats**
- Row 3 has **13 seats**
- ...
- Row 30 has **40 seats**

A **door prize** is awarded by:

1. Randomly selecting a row (each row equally likely)
 2. Randomly selecting a seat within that row
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Task 1: Probability that Seat 15 is Selected Given Row 20

$$P(S_{15} \mid R_{20}) = \frac{1}{30}$$

Task 2: Probability that Row 20 was Selected Given Seat 15

Using Bayes' rule:

$$P(R_{20} \mid S_{15}) = \frac{P(S_{15} \mid R_{20})P(R_{20})}{P(S_{15})}$$

Where:

$$P(S_{15}) = \sum_{k=1}^{30} P(S_{15} \mid R_k)P(R_k)$$

Numerical evaluation (as shown in lecture):

$$P(S_{15}) = 0.0342$$

Final result:

$$P(R_{20} \mid S_{15}) = \frac{\frac{1}{30} \cdot \frac{1}{30}}{0.0342} = 0.0325$$

8. Bayes' Theorem – Example 2: Communication System (Receiver RX)

A communication system sends **binary data** (0 or 1).

- The **receiver (RX)** may make mistakes:
 - A 0 may be detected as 1
 - A 1 may be detected as 0

The system is described using a **set of conditional probabilities**, and Bayes' theorem is used to compute posterior probabilities of transmitted symbols given received symbols.

(No further derivation shown on slides.)

End of Lecture 6

This scribe fully reconstructs **Lecture 6** exactly as taught, following the slides step by step and preserving all definitions, equations, and examples used in class.